



Day–Night Adaptation: A domain adaptation framework for medical image segmentation without source data

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ABSTRACT

Medical image segmentation models often suffer from performance degradation when deployed across clinical centers due to distribution shifts arising from differences in scanners, acquisition protocols, and patient populations. This challenge limits the robustness of deep models in real-world healthcare, where strong cross-center generalization is critical. Conventional unsupervised domain adaptation alleviates such shifts by jointly accessing source and target data, but this strategy is incompatible with strict privacy regulations that prohibit inter-center data sharing. More recent paradigms, including source-free domain adaptation (SFDA) and test-time adaptation (TTA), attempt to preserve privacy by relying only on target data. However, SFDA assumes a pre-collected target dataset prior to deployment, which is often impractical in clinical workflows, while TTA processes each case only once, underutilizing the accumulating data and risking unstable adaptation. To address these limitations, we propose **Day-Night Adaptation (DyNA)**, a clinically motivated, continuous adaptation framework that operates in synchrony with routine hospital schedules. During the day, DyNA performs adaptation on each test sample by learning a low-frequency prompt on the Fourier spectrum while keeping the backbone frozen, aided by a memory bank for initialization and a warm-up mechanism for stable optimization. At night, DyNA reuses the stored test samples collected during the day to conduct self-training with an improved teacher–student paradigm, where a global student bridges the gap between the EMA-updated teacher and the rapidly adapting student, enabling stable fine-tuning. In this way, DyNA unifies the advantages of online and offline adaptation, achieving reliable performance improvement without violating privacy constraints or requiring pre-collected target datasets. Extensive experiments on two benchmark medical image segmentation tasks demonstrate that DyNA consistently outperforms state-of-the-art SFDA and TTA methods. These results highlight the potential of DyNA as a privacy-preserving, deployment-oriented strategy for continual cross-center adaptation in real clinical environments. Codes and pre-trained source models will be publicly released at <https://github.com/Chen-Ziyang/DyNA>.

1. Introduction

Deep learning models have achieved remarkable success in medical image segmentation [1]. However, the development and deployment of segmentation models remain challenged by issues such as data scarcity [2], dataset bias [3], and distribution shift [4] across institutions. In this study, we mainly focus on the distribution shift, where models trained at one medical center (the source domain) fail to generalize to data from another (the target domain). These shifts

commonly arise from variations in imaging scanners, protocols, and operators. While unsupervised domain adaptation (UDA) techniques [5–10] can mitigate this problem by aligning source and target data distributions, they typically require concurrent access to data from both domains. This requirement often violates stringent patient privacy regulations and data-sharing agreements, restricting their practical utility in real-world clinical workflows.

To address these privacy constraints, source-free domain adaptation (SFDA) [11] and test-time adaptation (TTA) [12,13] have been developed. These paradigms adapt a pre-trained source model using

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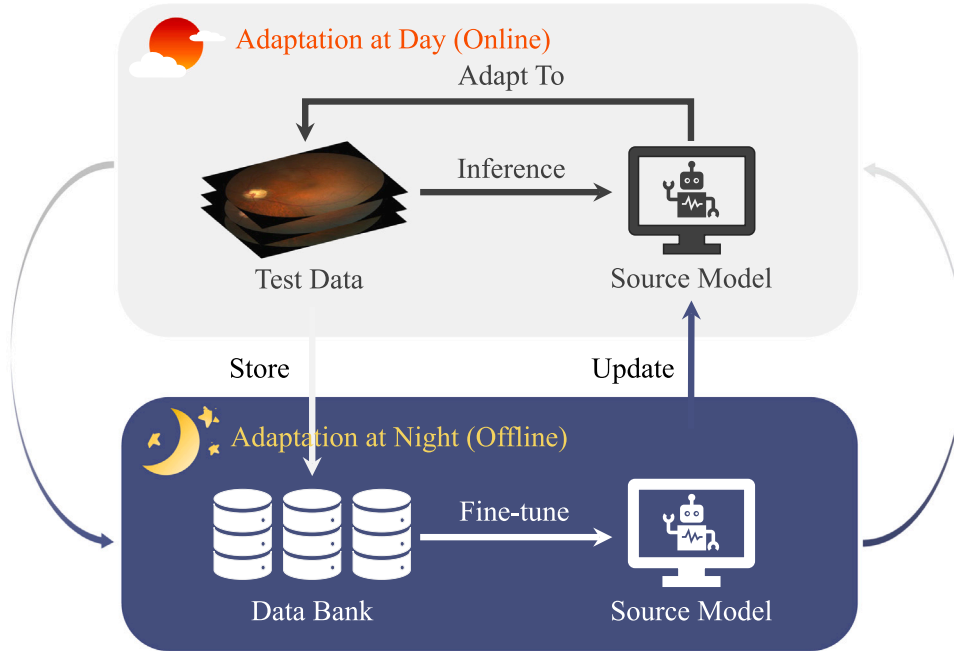


Fig. 1. Pipeline of our DyNA framework. We continuously perform adaptation through day-night cycles. During the day, the pre-trained source model performs inference and adaptation on the test data, and the test data is stored in a data bank. At night, the model's adaptation ability is further improved using the data bank.

only unlabeled target data, eliminating the need for source data access during adaptation. However, they present their own practical challenges. Conventional SFDA methods require a large, pre-collected target dataset for offline fine-tuning, a prerequisite that may not always be feasible before model deployment [14,15]. Conversely, TTA adapts the model online to each incoming test sample but discards the data after a single inference pass, thereby failing to leverage the accumulating target data for more robust, long-term adaptation (see Fig. 1 and Fig. 2).

In this work, we aim to address the limitations of both SFDA and TTA by exploiting a common operational pattern in clinical settings: patient data is acquired and processed during daytime working hours and then stored for future reference, while idle computing resources at night can be used for other purposes. Inspired by this workflow, we propose **Day-Night Adaptation (DyNA)**, a novel source-free framework that enables continuous model adaptation. DyNA operates in a cyclical manner similar to human memory consolidation. During the day, DyNA performs online test-time adaptation for each incoming patient sample, efficiently providing adapted segmentation results. The processed data is then stored in a data bank. During the night, the framework leverages the accumulated data to perform offline fine-tuning, consolidating the learned knowledge and preparing an improved model for the subsequent day. This iterative cycle allows the model to continuously improve its performance on the target domain without requiring a pre-collected dataset or access to source data. Unifying online and offline adaptation within this framework presents two primary challenges: preventing error accumulation during rapid daytime adaptation and ensuring stable, effective learning from limited, unlabeled data during nighttime fine-tuning. To overcome these hurdles, we introduce two key technical innovations. For daytime adaptation, we freeze the source model and train a lightweight, low-frequency prompt for each test sample to prevent the model from error accumulation. For nighttime fine-tuning, we introduce a global student model within a teacher-student paradigm to stabilize the training process and bridge the knowledge gap.

The contributions of this work are threefold.

- We develop a novel and clinically-motivated Day-Night Adaptation (DyNA) framework for continuous, source-free domain adaptation.
- We propose a prompt-based daytime strategy and a global student model-based nighttime strategy to ensure stable and effective adaptation.
- We simulate the day-by-day deployment under various domain shifts using public datasets from two benchmark medical image segmentation tasks. Extensive experiments demonstrate that DyNA significantly outperforms state-of-the-art TTA and SFDA method.

2. Related work

2.1. UDA methods

UDA methods aim to transfer knowledge from a labeled source domain to an unlabeled target domain. These approaches can be broadly classified by their level of domain alignment: input, feature, or output.

Input-level alignment methods transform the style of source-domain images to match the target domain while preserving semantic content. This is often achieved through generative models like CycleGAN [16] and its variants [17,18] or via frequency-domain manipulation [19,20]. For example, Wang et al. [17] introduced semantic consistency constraints in image translation to enhance segmentation consistency by leveraging both inter-domain and intra-domain consistency, while Yang et al. [19] proposed aligning domains by swapping low-frequency Fourier components between source and target images.

Feature-level alignment methods instead seek to learn domain-invariant representations by minimizing the distribution discrepancy in a shared feature space [5,6,8]. Sun et al. [6], for instance, used an orthogonality constraint to decompose features into domain-invariant and domain-specific components, which were then aligned using independent discriminators. Shin et al. [8] employed two independent encoders to disentangle intensity and non-intensity features, focusing adaptation on domain-invariant non-intensity features.

Output-level alignment methods typically employ adversarial learning in the model's output space to encourage the network to produce

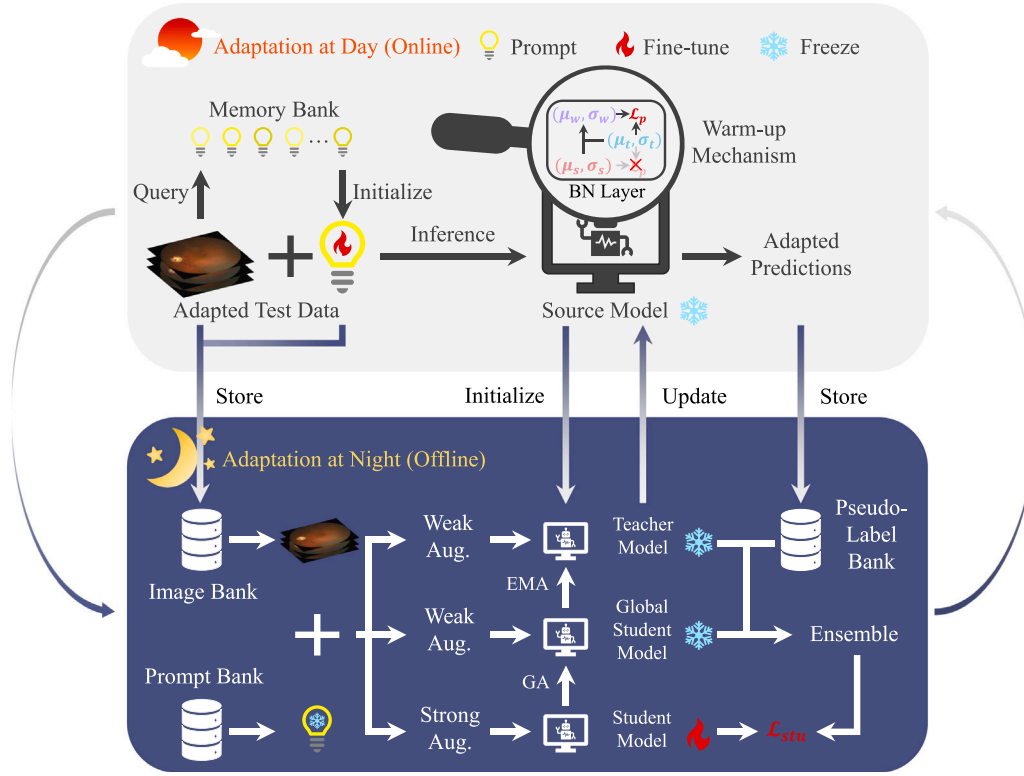


Fig. 2. Overview of the adaptation approaches utilized during day and night in our DyNA framework: (1) During the day, we freeze the pre-trained source model and train a learnable prompt for each test sample to boost adaptation performance. The test data, trained prompts, and adapted predictions are collected for use in the adaptation at night. μ_s/σ_s , μ_t/σ_t , and μ_w/σ_w denote the source, test, and warm-up statistics (mean/standard deviation), respectively. (2) At night, we freeze the trained prompts and fine-tune the source model using self-training on the collected data. A global student model is introduced to alleviate training instability. We integrate the pseudo-label and predictions of the global student model and teacher model to train the student model. The teacher model is updated by using the exponential moving average strategy and is subsequently frozen for inference the next day. ‘BN’: Abbreviation of ‘Batch Normalization’. ‘Aug.’: Abbreviation of ‘Augmentation’. ‘GA’: Abbreviation of ‘Global Average’. ‘EMA’: Abbreviation of ‘Exponential Moving Average’.

target predictions that are indistinguishable from source predictions [9, 10]. Wang et al. [9] utilized patch-based discriminators to generate segmentations with sharp boundaries and low entropy. Chen et al. [10] proposed a nuclear-norm Wasserstein discrepancy, integrated with the task classifier, to serve as a domain discriminator for adversarial learning.

While effective, all these UDA strategies require concurrent access to both source and target data, a condition that is often untenable in clinical settings due to data privacy concerns. Our work overcomes this limitation by operating in a source-free manner.

2.2. TTA methods

TTA adapts a pre-trained source model online during inference, using only the input test data [13]. A prominent line of TTA research employs self-supervised auxiliary losses to update the model on-the-fly [13,21–25]. These losses are often based on entropy minimization [23] to increase prediction confidence [13], sharpness-aware minimization to find a flat minima for model parameters [21], or dynamic learning rate adjustment based on feature similarity [22]. While these methods work well, directly updating the model makes them susceptible to error accumulation, particularly when supervision signals from the test data are noisy or unreliable. An alternative approach mitigates this issue by adapting only the batch normalization (BN) statistics instead of the full model [26–28]. By continuously adjusting the running mean and variance in BN layers, these methods modify feature representations to better suit the target domain while minimizing changes to the model [26,28]. However, the adaptation capacity of these methods will be limited due to minor modifications.

Our DyNA improves the model’s inference capability while preventing the model from error accumulation by learning a specific lightweight prompt for each test sample. Additionally, a fundamental limitation of existing TTA methods is their inefficient use of data, where each test sample is typically processed once and then discarded. Recent continual TTA studies have begun to model longer deployment streams more explicitly. Domain-consistency learning reduces catastrophic forgetting in semantic segmentation [29], mixture-of-incremental-experts supports stochastically interleaved medical domains [30], and stabilizing single-image continual adaptation by maintaining source-friendly target image and feature pools [31]. Nevertheless, these approaches primarily adapt models online along the test stream, whereas our DyNA framework stores test data acquired during the day and explicitly combines per-sample daytime adaptation with offline consolidation at night.

2.3. SFDA methods

SFDA methods fine-tune a pre-trained source model using only an unlabeled target-domain dataset, without requiring access to the original source data. These techniques generally fall into two categories: data-centric and self-training. **Data-centric methods** attempt to approximate the source distribution, either by generating a virtual source domain [32–34] or by identifying source-like samples within the target data [35–37]. For instance, Yang et al. [33] introduced a generator to synthesize source-style images from target data via Fourier-style mining, which were then used for domain alignment. Xia et al. [35] developed a contrastive category-wise matching module that enforces compactness within each category’s subspace using positive relations between target image pairs.

Self-training methods, inspired by semi-supervised learning [38, 39], leverage pseudo-labels [40,41] generated from the model's own predictions on target data. These approaches are often implemented within a teacher-student paradigm [42,43] to ensure stable learning and reduce error accumulation from noisy pseudo-labels. For example, Wu et al. [40] used a duplicated pre-trained decoder with perturbations to increase prediction diversity, providing more reliable pseudo-labels for adaptation. Li et al. [41] generated pseudo-labels from unlabeled target data, combining self-training with entropy minimization without requiring source data. Liu et al. [43] proposed periodically exchanging weights between the teacher and student models to integrate knowledge from past training iterations.

Despite their progress, a key limitation of existing SFDA methods is the requirement for a large pre-collected target dataset for offline fine-tuning. Our proposed framework avoids this demand by continuously collecting the data from daytime inference for subsequent nighttime fine-tuning.

3. Method

3.1. Problem definition and method overview

Our proposed framework, Day-Night Adaptation (DyNA), is designed to mirror the operational workflow of a clinical setting, where a pre-trained model must continuously adapt to new, unseen data from the local environment (see Fig. 2). Let the labeled source domain dataset be $\mathbb{D}^s = \{\mathcal{X}_i^s, \mathcal{Y}_i^s\}_{i=1}^{N^s}$ and the unlabeled target domain dataset be $\mathbb{D}^t = \{\mathcal{X}_i^t\}_{i=1}^{N^t}$, where $\mathcal{X}_i^* \in \mathbb{R}^{H \times W \times C}$ is the i th image with dimensions $H \times W$ and C channels, and $\mathcal{Y}_i^*$ is its corresponding class label. We define the segmentation model as $f: \mathcal{X} \rightarrow \mathcal{Y}$, where f_0 is pre-trained on $\mathbb{D}^s$ and deployed in the target domain. The DyNA framework operates in a continuous cycle of two distinct phases, which we describe here for the n th cycle.

Daytime Adaptation: This phase corresponds to a clinical standard operating hours. We employ our previous work [44] to perform adaptation. For each input test image $\mathcal{X}_i^t$, the model performs real-time, online adaptation to produce an immediate and accurate segmentation. To achieve this without compromising the stability of the core model f_n , we freeze its parameters and instead train a lightweight low-frequency prompt $P_i \in \mathbb{R}^{(\beta \times H) \times (\beta \times W) \times C}$. This prompt modifies the input image $\mathcal{X}_i^t$ to produce an adapted version $\tilde{\mathcal{X}}_i^t$ that better aligns with the source domain distribution. After inference, the raw image $\mathcal{X}_i^t$, its optimized prompt P_i , and the resulting prediction (pseudo-label) $\hat{\mathcal{Y}}_i$ are archived in a set of data banks, denoted as $S = \{\mathcal{X}_i^t, P_i, \hat{\mathcal{Y}}_i\}_{i=1}^{N^{test}}$.

Nighttime Fine-tuning: This phase corresponds to off-peak hours, leveraging idle computational resources to further improve the model. The goal is to consolidate the knowledge gained from the day's diverse test samples into the core model parameters. Using the data bank S , we fine-tune the model f_n via a improved self-training paradigm involving student, global student, and teacher models $f_n^{stu}, f_n^{glo}, f_n^{tea}$. After fine-tuning, the updated teacher model f_n^{tea} will be the new source model f_{n+1} for the next day. The data bank S is then cleared to prepare for the next cycle, preventing overfitting.

3.2. Daytime adaptation

During daytime inference, each test image $\mathcal{X}_i^t$ is adapted individually to mitigate the source-target domain shift before segmentation.

3.2.1. Low-frequency prompt

We hypothesize that domain shifts related to imaging hardware and protocols (e.g., lighting, contrast) primarily manifest in low-frequency components, while high-frequency details correspond more closely to semantic content. We therefore design a learnable prompt P_i that operates exclusively on the low-frequency amplitude of the image's Fourier spectrum. Let $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the Fast Fourier Transform (FFT)

and its inverse (IFFT) [45]. The learnable low-frequency prompt P_i of size $(\beta \times H) \times (\beta \times W) \times C$, where $\beta \in (0, 1)$, is padded with ones to match the image dimensions. It then modifies the low-frequency amplitude $\mathcal{F}(\mathcal{X}_i^t)$ while keeping the phase $\mathcal{F}_\phi(\mathcal{X}_i^t)$ unchanged. The adapted image $\tilde{\mathcal{X}}_i^t$ is obtained by:

$$\tilde{\mathcal{X}}_i^t = \mathcal{F}^{-1}([\text{PadOne}(P_i) \odot |\mathcal{F}(\mathcal{X}_i^t)|, \mathcal{F}_\phi(\mathcal{X}_i^t)]), \quad (1)$$

where $\text{PadOne}(\cdot)$ is the padding operator and $\odot$ is element-wise multiplication.

3.2.2. Prompt initialization

To ensure efficient and effective training, proper initialization of the learnable prompt for each test image is crucial. To this end, we construct a memory bank B that stores K pairs of keys and values $\{q_k, v_k\}_{k=1}^K$, where the keys q_k are the low-frequency components of previous test images, and the values v_k are their trained prompts. For the current test image $\mathcal{X}_i^t$, we extract its low-frequency component $|\mathcal{F}(\mathcal{X}_i^t)|$ and compute the cosine similarity between this component and each key q_k in the memory bank B as follows:

$$\text{Cos}(|\mathcal{F}(\mathcal{X}_i^t)|, q_k) = \frac{\langle |\mathcal{F}(\mathcal{X}_i^t)|, q_k \rangle}{\| |\mathcal{F}(\mathcal{X}_i^t)| \| * \|q_k\|}, \quad (2)$$

where $\langle \cdot, \cdot \rangle$ denotes the dot product, and $\| \cdot \|$ represents the Euclidean norm. We then rank the similarity scores and retrieve the values of the M most relevant keys from B to construct a support set $R_i = \{q_m, v_m\}_{m=1}^M$. The current prompt is initialized as a weighted sum of these values:

$$P_i = \sum_{m=1}^M \omega_m v_m, \quad (3)$$

where

$$\omega_m = \frac{e^{\text{Cos}(|\mathcal{F}(\mathcal{X}_i^t)|, q_m)}}{\sum_{j=1}^M e^{\text{Cos}(|\mathcal{F}(\mathcal{X}_i^t)|, q_j)}}. \quad (4)$$

The memory bank B is updated by the trained P_i using the First In First Out (FIFO) principle.

3.2.3. Prompt training

The prompt is optimized by aligning the batch normalization (BN) statistics of the adapted image with those of the source domain. However, direct alignment with static source statistics (μ_s, σ_s) can be unstable in the early adaptation process. We therefore introduce a warm-up mechanism that simulates the warm-up statistics (i.e., μ_w and σ_w) by dynamically combining the source and test statistics as follows:

$$\mu_w = \lambda \mu_t + (1 - \lambda) \mu_s, \quad \sigma_w = \lambda \sigma_t + (1 - \lambda) \sigma_s, \quad (5)$$

where $\lambda = \frac{1}{\sqrt{i/\tau+1}}$ smoothly decays from 1 to 0 as the test image index i increases, controlled by a temperature coefficient τ . The prompt is trained by minimizing the L1 distance between the warm-up and test statistics across all h BN layers:

$$\mathcal{L}_p = \frac{1}{h} \sum_h |\mu_w^h - \mu_t^h| + |\sigma_w^h - \sigma_t^h|, \quad (6)$$

where h denotes the index of BN layers. As inference progresses, the warm-up statistics gradually transition towards the source statistics, resulting in a gradual learning process.

3.2.4. Inference

After training the prompt P_i by minimizing the statistics alignment loss given in Eq. (6), the adapted image $\tilde{\mathcal{X}}_i^t$ is produced using trained P_i for the test image $\mathcal{X}_i^t$. The model then performs inference on $\tilde{\mathcal{X}}_i^t$ to obtain the adapted output $\hat{\mathcal{Y}}_i$, calculated as $\hat{\mathcal{Y}}_i = f_n(\tilde{\mathcal{X}}_i^t)$. The prediction $\hat{\mathcal{Y}}_i$ also serves as the pseudo-label at night. We summarize the overall process of daytime adaptation in Algorithm 1.

Algorithm 1: Daytime adaptation process of our DyNA.

Initialize : Source model f_n , initial prompt P_{init} filled with ones, empty memory bank B , empty set of data banks S , hyperparameters β , K , M , and τ .

1: **for** each time step i **do**

2: $\triangleright$ **Input:** current test image $\mathcal{X}_i^t$

3: # **Prompt Initialization:**

4: **if** $len(B) < M$ **then**

5: $P_i = P_{init}$

6: **else**

7: Construct the support set R_i from B . $P_i = \sum_{m=1}^M \omega_m v_m, v_m \in R_i$

8: **end if**

9: # **Prompt Training:**

10: Obtain $\tilde{\mathcal{X}}_i^t$ using P_i by Eq. (1)

11: Forward $f_n(\tilde{\mathcal{X}}_i^t)$ and calculate the warm-up statistics μ_w and σ_w by Eq. (5) to replace the μ_s and σ_s stored in f_n

12: Backward and Update P_i by $\mathcal{L}_p$ (Eq. (6))

13: # **Model Inference:**

14: Obtain $\tilde{\mathcal{X}}_i^t$ using updated P_i by Eq. (1). Forward $\hat{\mathcal{Y}}_i = f_n(\tilde{\mathcal{X}}_i^t)$ and store $\mathcal{X}_i^t$, P_i , and $\hat{\mathcal{Y}}_i$ into S

15: # **Memory Bank Updating:**

16: $B.enqueue(q_i, v_i)$, $q_i = |\mathcal{F}(\mathcal{X}_i^t)|$, $v_i = P_i$

17: **if** $len(B) > K$ **then**

18: $B.dequeue(q_{first}, v_{first})$

19: **end if**

20: $\triangleright$ **Output:** Adapted output $\hat{\mathcal{Y}}_i$

21: **end for**

22: $\triangleright$ **Output:** Set of data banks S

3.3. Nighttime fine-tuning

After daytime operations, the data bank set S (including a pseudo-label bank, an image bank, and a prompt bank) is used for offline model fine-tuning for further enhancement.

3.3.1. Self-training

We employ a teacher–student paradigm to mitigate error accumulation from noisy pseudo-labels. We initialize a student model f_n^{stu} , a global student model f_n^{glo} , and a teacher model f_n^{tea} the weights of the current source model f_n (i.e., $f_n^{stu} = f_n^{glo} = f_n^{tea} = f_n$). For each image in S , we first apply its corresponding prompt and generate the adapted image (as outlined in Eq. (1)), and then feed the strong augmented variant into the student model and weakly augmented variants into the teacher model and global student model. The weak augmentation strategy includes identity mapping, horizontal flipping, vertical flipping, and rotations of 90°, 180°, and 270°, while the strong augmentation strategy consists of brightness adjustment, contrast adjustment, Gamma transformation, Gaussian noise, and Gaussian blur.

To generate high-quality supervision, we combine the pseudo-label $\hat{\mathcal{Y}}$ with the predictions from the global student model and the teacher model (i.e., $\hat{p}^{glo}$ and $\hat{p}^{tea}$) as follows:

$$\mathbb{I} = \mathbb{I}((\hat{\mathcal{Y}} > T) \wedge (\hat{p}^{glo} > T) \wedge (\hat{p}^{tea} > T)) + \mathbb{I}((\hat{\mathcal{Y}} \leq T) \wedge (\hat{p}^{glo} \leq T) \wedge (\hat{p}^{tea} \leq T)) \quad (7)$$

where $\mathbb{I}$ denotes the indicator function, $\wedge$ represents the logical ‘and’ operation, and T is the threshold value.

The student model is then trained to match the masked predictions of the teacher, global student, and the original pseudo-label using a binary cross-entropy loss $\mathcal{L}_{bce}$:

$$\mathcal{L}_{stu} = \mathcal{L}_{bce}(\mathbb{I} * \hat{p}^{stu}, \mathbb{I} * \hat{p}^{glo}) + \mathcal{L}_{bce}(\mathbb{I} * \hat{p}^{stu}, \mathbb{I} * \hat{p}^{tea}) + \mathcal{L}_{bce}(\mathbb{I} * \hat{p}^{stu}, \mathbb{I} * \hat{\mathcal{Y}}), \quad (8)$$

where $\hat{p}^{stu}$ denotes the prediction of the student model, and $\mathcal{L}_{bce}$ represents the binary cross-entropy loss. By restricting supervision to reliable regions, this strategy effectively mitigates the impact of noisy pseudo-labels and stabilizes the self-training process.

3.3.2. Model update

The three models are updated using distinct strategies to ensure stable knowledge integration. After the student model f_n^{stu} is updated via $\mathcal{L}_{stu}$ at iteration r , the global student model f_n^{glo} is updated using a global average of the student model’s history. This maintains a highly stable, long-term representation of the learned knowledge:

$$f_{n,r}^{glo} = \frac{r * f_{n,r-1}^{glo} + f_{n,r}^{stu}}{r + 1}, \quad (9)$$

The teacher model f_n^{tea} is then updated via an exponential moving average (EMA) of the global student model’s weights. This allows for smooth and robust knowledge transfer from the stable global student to the teacher:

$$f_{n,r}^{tea} = \alpha * f_{n,r-1}^{tea} + (1 - \alpha) * f_{n,r}^{glo}, \quad (10)$$

where α is the EMA decay rate. After all training epochs, the final teacher model f_n^{tea} becomes the updated source model f_{n+1} for the next day’s clinical deployment. We summarize the overall process of nighttime fine-tuning in Algorithm 2.

Algorithm 2: Nighttime fine-tuning process of our DyNA.

Initialize : Source model f_n and hyperparameters T and α .

1: $\triangleright$ **Input:** Set of data banks S collected from daytime adaptation phase.

2: # **Model Initialization:** $f_{n,0}^{stu} = f_{n,0}^{glo} = f_{n,0}^{tea} = f_n$.

3: # **Model Update:**

4: **for** each iteration r **do**

5: **Update Student Model:** Update $f_{n,r}^{stu}$ by $\mathcal{L}_{stu}$

6: **Update Global Student Model:** Update $f_{n,r}^{glo}$ by Eq. (9)

7: **Update Teacher Model:** Update $f_{n,r}^{tea}$ by Eq. (10)

8: **end for**

9: Set $f_{n+1} = f_{n,r}^{tea}$ for the next adaptation loop and reset S

10: $\triangleright$ **Output:** Fine-tuned source model f_{n+1}

4. Experiments and results

4.1. Datasets and evaluation metrics

OD/OC Segmentation. Experiments for optic disc (OD) and optic cup (OC) segmentation were conducted on a composite dataset aggregated from five public sources, each representing a different domain: **Domain A** (RIM-ONE-r3 [46]), **Domain B** (REFUGE [47]), **Domain C** (ORIGA [48]), **Domain D** (REFUGE-Validation/Test [47]), and **Domain E** (Drishti-GS [49]). These domains contain 159, 400, 650, 800, and 101 images, respectively. Following the preprocessing protocol in [50], we extracted an 800×800 region of interest (ROI) centered on the OD from each image, which was then resized to 512×512 and normalized. Performance was evaluated using the Dice score coefficient (DSC).

Polyp Segmentation. For polyp segmentation, we used a dataset combining four public sources: **Domain A** (BKAI-IGH-NeoPolyp [51]), **Domain B** (CVC-ClinicDB [52]), **Domain C** (ETIS-LaribPolypDB [53]), and **Domain D** (Kvasir-Seg [54]), containing 1000, 612, 196, and 1000 images, respectively. As per the procedure in [55], all images were resized to 352×352 and normalized using ImageNet statistics. For this task, we report three evaluation metrics: DSC, the enhanced alignment metric (E_{ϕ}^{max}) [56], and the structural similarity metric (S_{β}) [57].

Data Partitioning. To evaluate competing TTA and SFDA methods, each domain’s dataset was partitioned into an 80:20 split. The 80% partition was used as the test set for all methods. The 20% partition was reserved exclusively as an unlabeled fine-tuning set for SFDA methods and was not accessible to TTA methods or our DyNA framework. To simulate continuous clinical deployment for DyNA, the 80% test set was further divided into periods of 10, 5, and 2 days, corresponding to processing 10%, 20%, and 50% of the test data per day-night cycle, respectively (see Table 1).

Table 1

Details of data split used for OD/OC segmentation task and polyp segmentation task. The training data is exclusively available for SFDA methods.

OD/OC Segmentation Task					
Domain	A (RIM-ONE-r3)	B (REFUGE)	C (ORIGA)	D (REFUGE-Validation/Test)	E (Drishti-GS)
Train/Test	32/127	80/320	130/520	160/640	20/81
Polyp Segmentation Task					
Domain	A (BKAI-IGH-NeoPolyp)	B (CVC-ClinicDB)	C (ETIS-LaribPolypDB)	D (Kvasir-Seg)	
Train/Test	200/800	122/490	39/157	200/800	

Table 2

Performance of our DyNA using different test data ratios, ‘Source Only’ baseline, four competing TTA methods, and four competing SFDA methods on the OD/OC segmentation task.

Methods		Domain A <i>DSC</i>	Domain B <i>DSC</i>	Domain C <i>DSC</i>	Domain D <i>DSC</i>	Domain E <i>DSC</i>	Average <i>DSC</i> ↑
Source Only (ResUNet-34)		67.14**	75.75**	74.54**	52.19**	68.57**	67.64**
TTA	DLTA [22]	74.32	77.79*	75.87**	55.96**	71.26**	71.04**
	DUA [28]	72.15**	75.82**	74.91**	56.86	72.48**	70.44**
	SAR [21]	74.28*	77.23**	74.87**	58.65	72.89**	71.58**
	DomainAdaptor [23]	74.95	76.54**	75.35**	56.79	72.02**	71.13**
SFDA	FSM [33]	74.14*	78.66	77.29**	56.16*	72.51**	71.75**
	UPL-SFDA [40]	74.36	76.39**	76.27**	58.82	69.07**	70.98**
	PETS [43]	70.74**	75.77**	74.88**	57.23	68.81**	69.49**
	PLPB [41]	72.72**	76.89**	77.14**	54.53**	74.74**	71.20**
DyNA (Ours)	10% (10 Days)	74.86	<u>78.93</u>	<u>79.58</u>	57.99	78.33	73.94
	20% (5 Days)	<u>75.36</u>	79.18	79.77	58.29	79.13	74.34
	50% (2 Days)	76.55	78.77	78.94	57.74	<u>78.57</u>	<u>74.12</u>

The best and second-best results in each column are highlighted in **bold** and underline, respectively. Statistical significance is evaluated using the Wilcoxon rank-sum test by comparing each competing method with DyNA (20%).

* Significance level of $p < 0.05$.

** Significance level of $p < 0.01$.

Table 3

Performance of our DyNA using different test data ratios, ‘Source Only’ baseline, and six competing methods on the polyp segmentation task.

Methods		Domain A			Domain B			Domain C			Domain D			Average		
		<i>DSC</i>	E_{ϕ}^{max}	S_a	<i>DSC</i>	E_{ϕ}^{max}	S_a	<i>DSC</i>	E_{ϕ}^{max}	S_a	<i>DSC</i>	E_{ϕ}^{max}	S_a	<i>DSC</i> ↑	E_{ϕ}^{max} ↓	S_a ↑
Source Only (PraNet)		75.84**	85.46**	82.97**	62.65**	78.09**	75.52**	74.10**	84.63*	81.44*	79.32**	88.10**	86.23**	72.98**	84.07**	81.54**
TTA	DLTA [22]	74.60**	85.45**	81.24**	59.86**	72.34**	75.21**	67.97**	80.38**	77.26**	66.93**	77.54**	79.46**	67.34**	78.93**	78.29**
	DUA [28]	77.09**	86.94**	83.49*	63.31**	76.94**	76.15**	71.77**	82.18**	80.56**	84.95**	92.50**	89.21*	74.28**	84.64**	82.35**
	SAR [21]	75.20**	86.04**	81.53**	59.80**	72.37**	74.96**	71.31**	82.72**	79.34**	66.10**	76.79**	78.60**	68.10**	79.48**	78.61**
	DomainAdaptor [23]	77.10**	87.24**	82.86**	66.61**	79.14**	79.18**	71.43**	82.31**	79.69**	74.35**	83.93**	83.13**	72.37**	83.16**	81.22**
SFDA	FSM [33]	<u>80.27</u>	89.44	85.39	66.47**	78.10**	76.54**	<u>77.71</u>	87.29	83.66	85.26**	93.15*	88.89**	77.43**	87.00**	83.62**
	UPL-SFDA [40]	79.45	88.26**	84.13	65.67**	76.94**	77.56**	77.76	86.71	<u>83.62</u>	82.40**	91.37**	86.93**	76.32**	85.82**	83.06**
	PETS [43]	76.67**	86.01**	83.45*	63.42**	78.95**	76.31**	74.63**	85.25*	81.92**	79.71**	88.13**	86.65**	73.61**	84.59**	82.08**
	PLPB [41]	78.19*	88.11	84.10	67.01**	80.61**	78.45**	74.80**	85.00**	81.95*	85.79*	93.61	89.42	76.45**	86.83**	83.48**
DyNA (Ours)	10% (10 Days)	79.92	89.23	84.83	73.86	86.69	82.73	76.41	86.20	82.81	86.63	94.03	90.06	79.20	89.04	85.11
	20% (5 Days)	80.46	89.67	<u>85.15</u>	75.19	87.93	83.36	76.67	86.38	82.91	<u>86.76</u>	94.18	90.10	79.77	89.54	85.38
	50% (2 Days)	80.26	<u>89.52</u>	85.02	<u>74.89</u>	<u>87.69</u>	<u>83.26</u>	76.15	86.03	82.57	86.79	<u>94.07</u>	90.14	<u>79.52</u>	<u>89.33</u>	<u>85.25</u>

The best and second-best results in each column are highlighted in **bold** and underline, respectively. Statistical significance is evaluated using the Wilcoxon rank-sum test by comparing each competing method with DyNA (20%).

* Significance level of $p < 0.05$.

** Significance level of $p < 0.01$.

4.2. Implementation details

Experimental Setup. We performed a comprehensive cross-domain evaluation. For each task, we trained the source model on each individual domain (source domain) using all source data and subsequently tested it on all remaining domains (target domains), where the mean metrics are computed on all target domains for evaluation. This resulted in a total of $4 \times 5 = 20$ adaptation scenarios for OD/OC segmentation and $3 \times 4 = 12$ scenarios for polyp segmentation.

Network Architectures. For OD/OC segmentation, we used a ResUNet [50] with a ResNet-34 [58] encoder. For polyp segmentation, we employed PraNet [55] with a Res2Net [59] backbone.

The encoder of ResUNet-34 is based on ResNet [58], composed of an initial convolutional layer and four convolutional blocks. The convolutional blocks contain 3, 4, 6, and 3 residual blocks, respectively.

Each residual block consists of two convolutional layers, each followed by a BN layer and ReLU activation. The decoder of ResUNet-34 includes five deconvolution layers, four skip connections, and a convolutional layer serving as the segmentation head.

The encoder of PraNet is based on Res2Net [59], similar to ResNet, but introduces a hierarchical residual-like layer between two convolutional layers within the residual block. This hierarchical layer splits features into smaller groups, which pass through a convolutional layer and are then integrated. The decoder of PraNet first aggregates features from the 2nd, 3rd, and 4th convolutional blocks into a global map using a parallel partial decoder. This global map is then upsampled and fed into three reverse attention modules along with these features, respectively. Each reverse attention module is followed by a segmentation head comprising four convolutional layers.

Hyperparameters in Daytime Adaptation. We performed adaptation within a single iteration for each test image (batch size of 1), a setting

Table 4

Results of ablation study on OD/OC segmentation task.

Methods							Do. A	Do. B	Do. C	Do. D	Do. E	Average
Day			Night			Test Data Ratio						
P	M	W	f^{glo}	$f^{tea} \leftarrow f^{glo}$	$f^{tea} \leftarrow f^{stu}$	10%/20%/50%	DSC	DSC	DSC	DSC	DSC	$DSC \uparrow$
✓						–	68.05**	76.10**	75.13**	52.80**	70.33**	68.48**
✓	✓					–	70.46**	77.38	76.28	54.35*	73.50**	70.39**
✓		✓				–	75.19	77.31	76.56	54.85	72.21**	71.22*
✓	✓	✓				–	74.27	78.09	76.87	55.92	75.93	72.22
✓	✓	✓			✓	10% (10 Days)	73.43*	78.96	76.65**	41.82**	73.48**	68.87**
✓	✓	✓	✓		✓	10% (10 Days)	74.33	79.08	78.88	56.75*	75.88**	72.98*
✓	✓	✓	✓	✓		10% (10 Days)	74.86	78.93	<u>79.58</u>	<u>57.99</u>	78.33	73.94
✓	✓	✓			✓	20% (5 Days)	76.12	78.53	75.17**	53.49**	74.36**	71.53**
✓	✓	✓	✓		✓	20% (5 Days)	<u>76.20</u>	<u>79.17</u>	78.37*	57.34	78.30	73.88
✓	✓	✓	✓	✓		20% (5 Days)	75.36	79.18	79.77	58.29	79.13	74.34
✓	✓	✓			✓	50% (2 Days)	75.56*	78.30	77.04**	56.88	76.30**	72.82*
✓	✓	✓	✓		✓	50% (2 Days)	76.03	78.98	77.66*	55.92**	78.26	73.37
✓	✓	✓	✓	✓		50% (2 Days)	76.55	78.77	78.94	57.74	<u>78.57</u>	<u>74.12</u>

The best and second-best results in each column are highlighted in **bold** and underline, respectively. ‘P’: Abbreviation of prompt. ‘M’: Abbreviation of memory bank. ‘W’: Abbreviation of warm-up. $f^{tea} \leftarrow f^*$ means utilizing f^* to update the teacher model. f^{glo} indicates introducing the global student model to provide predictions for better supervision information. Statistical significance is evaluated using the Wilcoxon rank-sum test. In the ablation study of daytime components, we compare each variant against the complete configuration (P+M+W) to determine significance. For the nighttime component ablation, significance tests are conducted between each variant and the setting of DyNA ($f^{glo} + f^{tea} \leftarrow f^{glo}$) under each test data ratio.

* Significance level of $p < 0.05$.

** Significance level of $p < 0.01$.

applied to both DyNA and all competing TTA methods for fair comparison. The Adam optimizer was used with a learning rate of 0.05 for OD/OC segmentation and 0.01 for polyp segmentation. Following the configurations in our previous work [44], the key hyperparameters for the prompt mechanism were set as follows: prompt size ratio $\beta = 0.01$, memory bank size $K = 40$, support set size $M = 16$, and warm-up temperature $\tau = 5$. Due to relatively complex structure of the PraNet decoder in the polyp segmentation task, we restricted the computation of the prompt loss $\mathcal{L}_p$ to only the BN layers within the encoder, rather than across the entire network.

Hyperparameters in Nighttime Fine-tuning. The model was fine-tuned for 10 epochs with a batch size of 4 using the SGD optimizer. The learning rate was set to 0.001 for the OD/OC task and 0.0003 for the polyp task. The supervision filtering threshold T was empirically set to 0.5. The EMA update rate α was set to 0.995 for the OD/OC task and 0.999 for the polyp task.

4.3. Comparison with state-of-the-art methods

We comprehensively compared our DyNA with the ‘Source Only’ baseline (source pre-trained model without adaptation), four competing TTA methods, and four competing SFDA methods. The competing TTA methods include an entropy-based method (SAR [21]), a method that utilizes a dynamic learning rate (DLTTA [22]), a method combining entropy minimization and statistics fusion (DomainAdaptor [23]), and a method that modifies the BN statistics (DUA [28]). For SFDA, we compared a data-centric approach (FSM [33]) and three self-training methods (UPL-SFDA [40], PETS [43], and PLPB [41]), with PETS utilizing a teacher–student paradigm for self-training. For a fair comparison, all methods were reimplemented using the same experimental setup and backbones.

4.3.1. OD/OC segmentation results

As shown in Table 2, all adaptation methods successfully improved upon the ‘Source Only’ baseline, confirming the necessity of addressing domain shift. Both TTA and SFDA methods achieved comparable performance. Our DyNA framework, however, consistently outperformed all competing methods across most scenarios and the overall performance with a significance level of $p < 0.01$, achieving the superior average DSC. This result highlights the benefit of the cyclical day-night adaptation strategy. Furthermore, DyNA’s performance remained

robust irrespective of the daily data partition, a stability we attribute to the error-mitigating frozen-model approach during the day and the stable global student model during the night.

4.3.2. Polyp segmentation results

Results for polyp segmentation are presented in Table 3. In this more challenging task, SFDA methods generally surpassed TTA methods, indicating that access to unlabeled target domain data for fine-tuning is beneficial. Notably, several entropy-minimization-based TTA methods (DLTTA, SAR, DomainAdaptor) underperformed the ‘Source Only’ baseline, suggesting that naive test-time optimization can be detrimental when faced with significant appearance variations. In contrast, DyNA delivers state-of-the-art performance in almost all scenarios and significantly outperforms other competing methods on the overall performance across all metrics. This demonstrates that by effectively reusing test data, our framework can achieve robust adaptation without requiring a pre-collected unlabeled dataset. Similar to the results on the OD/OC segmentation task, our DyNA maintains robust adaptation performance across varying test data ratios.

4.4. Ablation study

We conducted ablation studies on the OD/OC segmentation task to analyze the contributions of DyNA’s core components (Table 4).

Daytime Adaptation: The low-frequency prompt alone provided a marginal benefit. Performance was substantially improved by introducing the memory bank for prompt initialization and further enhanced by the warm-up mechanism. The complete daytime configuration, integrating all three components, achieved the best results, demonstrating their synergistic effect on stable and effective online adaptation.

Nighttime Fine-tuning: We analyzed the effect of the global student model and the choice of model for the teacher update. The introduction of the global student model as an additional source of supervision for the student model consistently improved performance. Furthermore, using the stable global student model to update the teacher via EMA yielded superior results compared to updating from the student model directly. This performance gain was particularly pronounced in long-term adaptation scenarios (i.e., 10% daily test data ratio), underscoring the global student model’s role in maintaining stability over extended deployment.

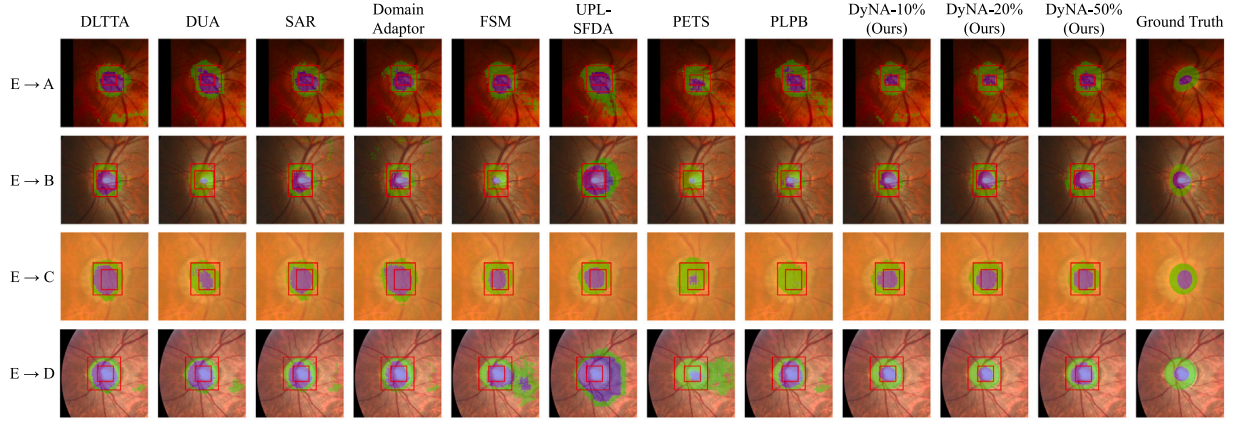


Fig. 3. Visualization of OD/OC segmentation results obtained by four competing TTA methods, four competing SFDA methods, and our DyNA. From top to bottom: segmentation results and ground truth obtained from the scenario denoted as “source → target”. From left to right: segmentation results of the TTA methods (DLTTA, DUA, SAR, and DomainAdaptor), SFDA methods (FSM, UPL-SFDA, PETS, and PLPB), and our DyNA using different test data ratios, followed by the ground truth. To highlight potential over- or under-segmentation, the ground truth bounding boxes of OD and OC are overlaid on each segmentation result. Best viewed in color.

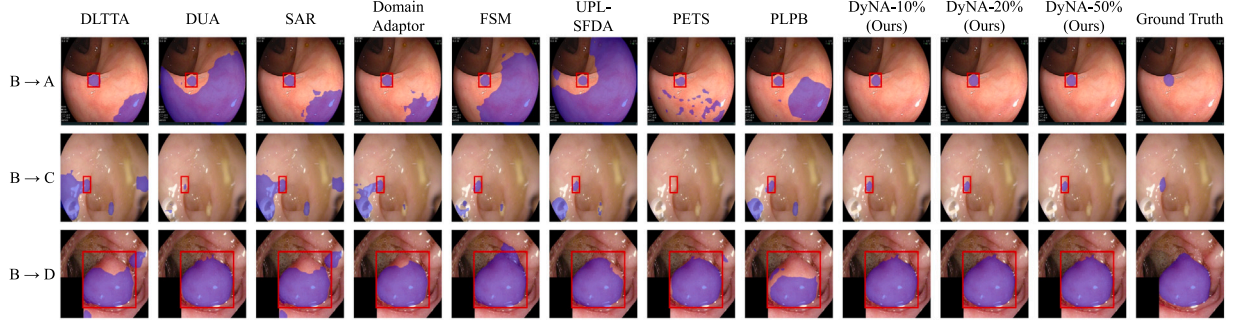


Fig. 4. Visualization of polyp segmentation results obtained by four competing TTA methods, four competing SFDA methods, and our DyNA. From top to bottom: segmentation results and ground truth obtained from the scenario denoted as “source → target”. From left to right: segmentation results of the TTA methods (DLTTA, DUA, SAR, and DomainAdaptor), SFDA methods (FSM, UPL-SFDA, PETS, and PLPB), and our DyNA using different test data ratios, followed by the ground truth. To highlight potential over- or under-segmentation, the ground truth bounding boxes of foreground are overlaid on each segmentation result. Best viewed in color.

Table 5

Results of various combinations of our DyNA and other TTA/SFDA methods using different test data ratios on the OD/OC segmentation task. The red numbers indicate the performance gain achieved by the combination over the method without our DyNA.

Methods			Do. A	Do. B	Do. C	Do. D	Do. E	Average
Day	Night	Test Data Ratio						
		10%/20%/50%	DSC	DSC	DSC	DSC	DSC	DSC ↑
DLTTA [22]	–	–	74.32	77.79	75.87	55.96	71.26	71.04
DLTTA [22]	DyNA (Ours)	10% (10 Days)	76.98	78.64	78.11	59.76	71.78	73.05 +2.01
		20% (5 Days)	76.54	78.79	77.66	58.63	72.10	72.74 +1.70
		50% (2 Days)	75.30	78.13	76.89	56.77	71.71	71.76 +0.72
SAR [21]	–	–	74.28	77.23	74.87	58.65	72.89	71.58
SAR [21]	DyNA (Ours)	10% (10 Days)	75.58	77.27	77.03	61.65	73.20	72.95 +1.37
		20% (5 Days)	75.66	77.31	77.03	61.80	73.13	72.99 +1.41
		50% (2 Days)	74.97	77.24	76.08	59.39	73.47	72.23 +0.65
–	FSM [33]	10% (10 Days)	75.06	77.49	76.47	56.05	71.88	71.39
		20% (5 Days)	74.02	77.25	76.29	55.37	71.86	70.96
		50% (2 Days)	71.38	76.84	75.57	53.92	71.59	69.86
DyNA (Ours)	FSM [33]	10% (10 Days)	75.27	78.29	76.62	57.47	75.81	72.69 +1.30
		20% (5 Days)	75.14	78.23	76.61	57.13	75.36	72.49 +1.53
		50% (2 Days)	74.77	78.20	76.71	56.53	75.17	72.28 +2.42
–	PLPB [41]	10% (10 Days)	71.54	77.29	75.06	53.21	70.59	69.54
		20% (5 Days)	71.19	76.42	75.16	52.99	70.21	69.20
		50% (2 Days)	69.50	75.81	74.82	51.62	67.80	67.91
DyNA (Ours)	PLPB [41]	10% (10 Days)	73.42	78.28	76.46	56.08	76.18	72.08 +2.54
		20% (5 Days)	73.61	78.46	76.37	55.83	75.48	71.95 +2.75
		50% (2 Days)	73.03	78.29	76.28	56.64	75.91	72.03 +4.12

Table 6Results of setting α to various values on the OD/OC segmentation task.

Methods		Domain A	Domain B	Domain C	Domain D	Domain E	Average
α	Test Data Ratio						
	10%/20%/50%	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i> $\uparrow$
0.99	10% (10 Days)	70.63	78.42	79.50	56.91	77.05	72.50
	20% (5 Days)	75.86	78.99	78.93	59.14	77.70	74.12
	50% (2 Days)	75.92	78.67	78.42	57.55	78.11	73.73
0.995	10% (10 Days)	74.86	78.93	<u>79.58</u>	57.99	78.33	73.94
	20% (5 Days)	75.36	79.18	79.77	<u>58.29</u>	79.13	74.34
	50% (2 Days)	76.55	78.77	78.94	57.74	<u>78.57</u>	<u>74.12</u>
0.999	10% (10 Days)	75.85	78.88	79.11	56.18	77.57	73.52
	20% (5 Days)	<u>76.35</u>	78.94	78.53	57.70	78.15	73.93
	50% (2 Days)	76.08	78.60	78.17	57.39	77.67	73.58

The best and second-best results in each column are highlighted in **bold** and underline, respectively.**Table 7**Results of setting α to various values on the polyp segmentation task.

Methods		Domain A			Domain B			Domain C			Domain D			Average		
α	Test Data Ratio															
	10%/20%/50%	<i>DSC</i>	E_{ϕ}^{max}	S_a	<i>DSC</i>	E_{ϕ}^{max}	S_a	<i>DSC</i>	E_{ϕ}^{max}	S_a	<i>DSC</i>	E_{ϕ}^{max}	S_a	<i>DSC</i> $\uparrow$	E_{ϕ}^{max} $\uparrow$	S_a $\uparrow$
0.995	10% (10 Days)	75.99	84.37	80.42	62.54	<u>77.74</u>	77.01	64.76	77.18	76.75	77.65	87.07	84.93	70.23	81.59	79.78
	20% (5 Days)	75.05	84.20	79.98	71.19	85.29	80.63	63.19	74.46	70.81	82.44	90.81	87.40	72.97	83.69	79.70
	50% (2 Days)	79.53	89.23	84.49	73.82	86.38	82.61	63.01	75.53	74.33	85.42	93.13	89.06	75.44	86.06	82.62
0.999	10% (10 Days)	79.92	89.23	84.83	73.86	86.69	82.73	76.41	86.20	82.81	86.63	94.03	90.06	79.20	89.04	85.11
	20% (5 Days)	80.46	89.67	<u>85.15</u>	<u>75.19</u>	<u>87.93</u>	<u>83.36</u>	76.67	86.38	<u>82.91</u>	86.76	94.18	90.10	79.77	89.54	85.38
	50% (2 Days)	<u>80.26</u>	<u>89.52</u>	<u>85.02</u>	74.89	87.69	83.26	76.15	86.03	82.57	<u>86.79</u>	94.07	<u>90.14</u>	79.52	89.33	85.25
0.9995	10% (10 Days)	79.39	88.81	84.51	74.35	87.09	83.04	76.31	86.01	82.82	86.65	94.09	90.07	79.18	89.00	85.11
	20% (5 Days)	79.83	89.15	84.79	75.24	87.87	83.48	<u>76.60</u>	<u>86.24</u>	83.00	86.83	94.23	90.15	<u>79.63</u>	<u>89.38</u>	<u>85.36</u>
	50% (2 Days)	79.83	89.14	84.77	75.08	87.94	83.34	76.36	86.02	82.88	86.73	94.04	90.05	79.50	89.28	85.26

The best and second-best results in each column are highlighted in **bold** and underline, respectively.**Table 8**

Results of employing different models for next day's deployment on the OD/OC segmentation task.

Methods		Domain A	DomainB	DomainC	DomainD	DomainE	Average
Model	Test Data Ratio						
	10%/20%/50%	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i> $\uparrow$
f^{sta}	10% (10 Days)	68.29	73.58	76.41	39.62	73.00	66.18
	20% (5 Days)	71.03	78.37	77.99	49.82	77.01	70.85
	50% (2 Days)	75.51	78.70	77.70	57.17	78.03	73.42
f^{glo}	10% (10 Days)	75.75	79.03	78.38	52.89	68.31	70.87
	20% (5 Days)	75.23	<u>79.15</u>	77.60	57.92	78.45	73.67
	50% (2 Days)	<u>76.25</u>	78.80	78.37	56.88	77.80	73.62
f^{tea}	10% (10 Days)	74.86	78.93	<u>79.58</u>	<u>57.99</u>	78.33	73.94
	20% (5 Days)	75.36	79.18	79.77	58.29	79.13	74.34
	50% (2 Days)	76.55	78.77	78.94	57.74	<u>78.57</u>	<u>74.12</u>

The best and second-best results in each column are highlighted in **bold** and underline, respectively.

4.5. Qualitative results

Qualitative results are presented for representative ‘source $\rightarrow$ target’ scenarios in Fig. 3 (OD/OC) and Fig. 4 (Polyp). To facilitate comparison, ground truth boundaries are overlaid on all segmentation outputs. The visualizations show that DyNA produces segmentations with significantly less over- and under-segmentation compared to competing methods. For polyp segmentation (Fig. 4), DyNA demonstrates robust performance on anatomies of varying scales, accurately segmenting both small (rows 1–2) and large (row 3) polyps where other methods often fail. These visual results corroborate the quantitative findings and highlight the practical efficacy of the proposed day-night strategy.

5. Discussion

5.1. Extensibility of the DyNA framework

To demonstrate the modularity of our day-night paradigm, we conducted experiments where the core components of DyNA were replaced with existing adaptation methods. Specifically, we substituted our daytime prompt-based adaptation with established TTA methods (DLTTA and SAR) and our nighttime self-training with SFDA methods (FSM and PLPB).

The results, presented in Table 5, show that integrating these methods into the DyNA cycle consistently improves their baseline performance across all test data ratios. A key observation is that while standalone SFDA methods exhibit performance degradation in long-term

Table 9

Performance of our DyNA with a data ratio of 50% on the OD/OC segmentation task in a long-term continual adaptation scenario.

Round	Domain A	Domain B	Domain C	Domain D	Domain E	Average
	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i>	<i>DSC</i> ↑
1	73.72	76.92	77.91	55.11	79.29	72.59
2	71.02	76.31	78.63	54.82	80.26	72.21
3	65.71	75.35	78.34	54.02	81.18	70.92
4	52.54	74.39	77.78	53.34	80.12	67.63
5	43.28	73.41	76.18	53.06	79.31	65.05
6	36.59	72.89	74.68	52.99	78.38	63.11
7	35.65	72.51	73.50	52.98	76.99	62.33
8	35.12	72.18	72.55	53.00	74.67	61.50
9	34.62	71.85	71.75	53.01	71.35	60.52
10	34.15	71.76	71.16	53.03	66.68	59.42

Table 10

Complexity results of our DyNA and other competing methods on the OD/OC segmentation task for the scenario (Source: Domain A → Target: Domain B).

Methods	Test Phase			Training Phase (only for SFDA and DyNA)		
	Inference Time per Image (s)	Learnable Parameters	GFLOPs	Total Time (s)	Learnable Parameters	GFLOPs
Source Only (ResUNet-34)	0.02	–	80.37	–	–	–
TTA	DLTTA [22]	0.09	19 136	160.76	–	–
	DUA [28]	0.04	–	160.76	–	–
	SAR [21]	0.11	11 968	241.14	–	–
	DomainAdaptor [23]	0.61	19 136	160.76	–	–
SFDA	FSM [33]	0.02	–	80.37	127.69	22.55 M
	UPL-SFDA [40]	0.02	–	80.37	229.45	24.83 M
	PETS [43]	0.02	–	80.37	158.71	45.10 M
	PLPB [41]	0.02	–	80.37	163.62	22.55 M
DyNA (Ours)	10% (10 Days)	0.12	75	160.76	476.78	22.55 M
	20% (5 Days)	0.12	75	160.76	343.59	22.55 M
	50% (2 Days)	0.12	75	160.76	187.50	22.55 M

deployment scenarios (i.e., with smaller daily data volumes), incorporating them into DyNA’s nighttime phase stabilizes and enhances their performance. This suggests that the iterative, cyclical nature of our framework provides a robust strategy for lifelong learning, effectively leveraging new data to refine the model while mitigating catastrophic forgetting.

5.2. Sensitivity to EMA hyperparameter α

We analyzed the sensitivity of DyNA to the hyperparameter α , which controls the exponential moving average (EMA) update rate for the teacher model. As shown in Tables 6 and 7, the impact of α varies between the two segmentation tasks.

For OD/OC segmentation, DyNA exhibited high robustness, with performance remaining stable across a range of α values in nearly all scenarios. In contrast, for the more challenging polyp segmentation task, performance was more sensitive to this parameter. While the model was robust to slower update rates, a faster update (a higher α) led to performance degradation. This finding suggests that for tasks with high intra-domain variance like polyp segmentation, a more cautious and gradual integration of knowledge into the teacher model is critical for stable learning. Our experiments confirmed that setting α to 0.995 for OD/OC segmentation and 0.999 for polyp segmentation yielded the best overall performance.

These results further suggest a practical strategy for deployment on a new task: when the target-domain characteristics are unknown, a conservative value ($\alpha = 0.999$) can be used as the default choice to reduce the risk of error accumulation. If the anatomical structures are relatively stable, visually well defined, and less variable across cases, α can be slightly reduced to enable faster teacher updates.

5.3. Optimal model for next-cycle deployment

Our nighttime phase maintains three distinct models: the student, the global student, and the teacher. We performed an experiment on

the OD/OC segmentation task to identify which model serves as the optimal foundation for the next day’s deployment.

The results, summarized in Table 8, unequivocally show that deploying the teacher model consistently achieves the best performance across all scenarios. Using the student model, which is subject to rapid updates on small batches of data, can lead to significant performance degradation, particularly in long-term scenarios where the daily test data ratio is low. The global student model offers more stability but is still outperformed by the teacher. This outcome validates our EMA update strategy, where the teacher model functions as a stabilized, momentum-driven version of the global student, effectively filtering high-frequency fluctuations from the student’s training trajectory to provide a robust and reliable model for subsequent clinical inference.

5.4. Long-term continual adaptation

We conducted a long-term continual adaptation experiment on the OD/OC segmentation task using DyNA with a data ratio of 50%. In this setting, DyNA is deployed to a series of target domains for 10 rounds, where the target distributions change continuously and previously encountered domains will reappear in the next round. As shown in Table 9, the performance of DyNA continuously declines on Domains A, C, and E, but tends to converge on Domains B and D. These results suggest that DyNA is able to retain useful adaptive knowledge for some domains; however, it cannot completely prevent forgetting. Therefore, achieving universally stable performance under long-term domain transitions remains an open challenge.

5.5. Complexity analysis

We further evaluate the computational complexity of DyNA and other competing methods by reporting both test and training overhead on both segmentation tasks. The detailed results are summarized in Tables 10 and 11. At test phase, DyNA introduces an acceptable increase in latency, requiring 0.12 s/image for OD/OC segmentation and

Table 11

Complexity results of our DyNA and other competing methods on the polyp segmentation task for the scenario (Source: Domain A → Target: Domain B).

Methods		Test Phase			Training Phase (only for SFDA and DyNA)		
		Inference Time per Image (s)	Learnable Parameters	GFLOPs	Total Time (s)	Learnable Parameters	GFLOPs
Source Only (PraNet)		0.04	–	26.19	–	–	–
TTA	DLTTA [22]	0.18	66 218	52.38	–	–	–
	DUA [28]	0.08	–	52.38	–	–	–
	SAR [21]	0.25	41 098	78.56	–	–	–
	DomainAdaptor [23]	1.28	66 218	52.38	–	–	–
SFDA	FSM [33]	0.04	–	26.19	161.61	32.55 M	680.89
	UPL-SFDA [40]	0.04	–	26.19	262.52	49.62 M	628.52
	PETS [43]	0.04	–	26.19	258.01	65.10 M	628.52
	PLPB [41]	0.04	–	26.19	219.17	32.55 M	419.01
DyNA (Ours)	10% (10 Days)	0.25	27	52.38	642.02	32.55 M	523.76
	20% (5 Days)	0.25	27	52.38	513.43	32.55 M	523.76
	50% (2 Days)	0.25	27	52.38	312.10	32.55 M	523.76

0.25 s/image for polyp segmentation. Although this additional cost is higher than that of the source-only model and several SFDA methods, it remains comparable to most TTA-based approaches and is substantially lower than that of DomainAdaptor. This favorable efficiency can be attributed to the fact that DyNA updates only a lightweight prompt module, involving merely 75 parameters for OD/OC segmentation and 27 parameters for polyp segmentation, while keeping the model fixed. For the training phase, DyNA requires an additional nighttime adaptation procedure. Nevertheless, its computational cost remains well controlled. In particular, the overall GFLOPs of DyNA are consistently comparable to, or lower than, other SFDA methods across both tasks. This indicates that the improved adaptation capability of DyNA is not achieved at the expense of excessive computational burden. Overall, these results suggest that DyNA provides a favorable trade-off between computational efficiency and adaptation performance, which is desirable for real-world deployment.

6. Conclusion

In this paper, we propose DyNA, a novel source-free framework that enables the continuous adaptation of a pre-trained segmentation model to better align with real-world clinical workflows and fully utilize input test data. Our method operates in a cycle that mimics clinical operations: a daytime phase performs efficient, per-image adaptation using a lightweight, low-frequency prompt while keeping the core model frozen, and a nighttime phase uses the accumulated data to fine-tune the model via a stable self-training paradigm featuring a global student model. In this study, we constructed simulation-based evaluations using public datasets to approximate day-by-day deployment under various domain shifts. Extensive experiments on two medical image segmentation tasks demonstrated that DyNA significantly outperforms existing TTA and SFDA methods across a variety of scenarios. Furthermore, we showed that the DyNA paradigm is a flexible and extensible framework that can be combined with other adaptation techniques to enhance their performance. This work represents a step towards building truly adaptive medical AI systems that can evolve and improve over time as they are exposed to new data, directly addressing the critical challenge of model generalization in clinical practice.

There are also several limitations within this work. First, the proposed DyNA assumes a practical day–night adaptation workflow with temporary target-domain data buffering and additional computational resources for nighttime updates, which may limit its flexibility. Second, the effectiveness of the nighttime stage depends on the quality of pseudo labels. Third, the current study is restricted to regular 10-, 5-, and 2-day schedules and 2D segmentation tasks. Future work will aim to improve the flexibility and robustness of DyNA in diverse real-world clinical deployment settings. We will extend the framework to more realistic clinical data streams and 3D medical image segmentation tasks, and further evaluate it in longitudinal single-hospital deployment

scenarios where domain shifts may evolve gradually over time. Moreover, we will explore mechanisms for mitigating pseudo-label noise, such as experience replay to retain representative historical samples and Elastic Weight Consolidation to constrain updates of important model parameters.

CRediT authorship contribution statement

Ziyang Chen: Writing – review & editing, Writing – original draft, Validation, Methodology. **Yiwen Ye:** Visualization, Data curation. **Yongsheng Pan:** Writing – review & editing, Supervision, Project administration. **Jingfeng Zhang:** Resources, Data curation. **Yong Xia:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to polish the language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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